Practical Work

Summary

This assignment aims to give students the opportunity to apply the image processing and analysis concepts covered in class. Students will develop a program in C or C++ that operates on the video provided for this practical assignment, in order to obtain the desired information (see below). This work will culminate in the submission of the developed source code, in addition to the mandatory oral defense of the work carried out.

Completion of practical work

The file "**VC-TP.zip**" contains a description of the work and its objectives, and includes**:**

- Video to use;
- Source code with an example of reading/capturing and displaying video using the OpenCV library;
- Excerpt from European Union legislation on the classification of citrus fruits.

The work should be developed in groups, and the **groups should consist of 4 members**. However, **Erasmus students may form groups of 2 members**.

Students are expected to use the knowledge acquired throughout the Computer Vision classes and develop the functionalities necessary to achieve the objectives.

Note that all code must be written in C or C++, and students may use (in addition to the OpenCV functions identified in the "CodigoExemplo.cpp" file, or similar functions depending on the OpenCV version) **up to 3 more functions or instances of classes from the OpenCV library**. For example, the following line of code would reach this limit, using `cv::rectangle()`, `cv::Point()`, and `cv::Scalar()`:

```
cv::rectangle(frame, cv::Point(10,2), cv::Point(100,20), cv::Scalar(255,255,255), -1);
```

The use of the following is not permitted: external image processing libraries not covered in class; code available in repositories (such as GitHub); as well as any other code whose authorship does not belong to members of the group.

The work must also be defended by all group members in an **oral presentation** , with **questions about the work being posed** to the various group members. No report or PowerPoint presentation is required. The defense should focus on the strategy followed and its implementation (source code), the difficulties encountered, and the solutions found. Students are expected to master all the source code used by the group to complete the practical work .

Objectives

The groups should develop a program in C or C++ that processes the video "**video.avi**" to obtain the following set of information in each *frame* of the video:

- Total number of oranges (total count, from the first *frame* of the video to the current *frame*);
- Number of oranges detected in the current *frame;*
- Area and perimeter of each orange;
- Location (bounding area) and center of gravity of each orange;
- Size of oranges (see annex "Quality Standards for Citrus Fruits", section "III. Size");
- Category of each orange (see annex "Common Quality Standards for Citrus Fruits", section "II. Quality Requirements", point "D. Classification").

This information should be displayed in real time over the provided video.

Also keep in mind that:

- The video has a resolution of 1280x720;
- The video has a *frame rate* of 25 fps ;
- 280 pixels correspond to 55mm.



Figure: Suggested way to present information in the video:

Assessment

The following factors will be used as evaluation criteria:

- Program quality:
  - Development of the functionalities described in the assignment statement;
  - level of optimization of the implemented functions;
  - The program is functioning correctly;
  - added value [1].
- Quality of the code and related comments, as well as the oral presentation:
  - A correct and complete description of the program's structure;

[1]Added value refers to how the work stands out (positively) from the rest.

- Description of the techniques applied in the development of the functionalities .

- Respect the rules for submitting the work.
- The level of knowledge that the student possesses about the work being done.

The collective nature of carrying out group work does not preclude the fact that the evaluation is individual for each member of the group.

Each member of the group must demonstrate knowledge of all the submitted code.

IMPORTANT NOTES:

- The defense of the work is mandatory. Failure to do so will result in a grade of 0 (zero) for the practical component.
- For the defense, on the scheduled day, the group must bring at least one PC with the editable and compilable code. However, it is recommended that they bring more than one computer to guard against any potential problems.
- The source code for the presentation/defense must be the same as the one submitted for evaluation, but **all comments must be removed** beforehand.

Deadlines

Completing the assignment requires submitting the source code files in digital format. Do not submit **the** project folder, only the source code files (**`.c`**/**`.h`** or **`.cpp`**/**`.h`**).

The work must be submitted to the instructor via the submission *link available on the Moodle page for this course, by the date and time specified (also available on the course's Moodle page).*

The submission of the practical work must **strictly adhere to** the following requirements:

- The source code files should be placed in a zip file named "**VC-GY-xxxx-xxxx-xxxx-xxxx.zip**", where:
  - **xxxx** should be filled in with the student number of each member of the group);
  - **GY** should be filled with **G1** for group 1, **G2** for group 2, etc.
- Only 1 (one) member from each group should submit the work.
- The zip file name should only include the student numbers of those who actually completed the work and will therefore be subject to evaluation.

The deadline for submissions is **May 18th** at **11:00 PM**. Submissions **made after this date will not be considered**.

Ethical conduct

Lack of transparency in assessments, whether in person or not, is naturally illegal and immoral. All sources used to support assignments must be clearly and mandatorily referenced. Any plagiarism, copying, or improper academic conduct will be penalized with the annulment of the work. If it is found that there are demonstrably similar works (where, for example, only the names of variables from another code have been changed) between groups, all similar works will be annulled .